Linear algebra: Exercise session 3

Morteza Rahimi

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1. In a garden there are only rabbits and chickens. After counting there appear to be 108 legs and 39 heads. How many chickens and rabbits are in the garden?

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2. Describe the set of solutions of the following systems by Gauss-Jordon elimination algorithm:

(a) $\begin{cases} 3x_1 + 2x_2 = 5, \\ 6x_1 + 3x_2 = 7, \end{cases}$

(b) $\begin{cases} 3x_1 + 2x_2 + 4x_3 = 5, \\ x_1 + x_2 - 3x_3 = 2, \\ 4x_1 + 3x_2 + x_3 = 7, \end{cases}$

(c) $\begin{cases} x_1 + 2x_2 + x_3 = 0, \\ x_1 - x_2 + 2x_3 = 1, \\ 3x_1 + 5x_3 = 3, \end{cases}$

(d) $\begin{cases} 2x_1 + 3x_2 + x_3 = 1, \\ x_1 + x_2 + x_3 = 3, \\ 3x_1 + 4x_2 + 2x_3 = 4, \end{cases}$

(e) $\begin{cases} x_1 + 2x_2 + x_3 = 11, \\ 2x_1 + x_2 - x_3 = 1, \\ x_1 - x_2 + 3x_3 = 10, \end{cases}$

(f) $A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 7 & 5 \\ 0 & -2 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 8 \\ 20 \\ 0 \end{bmatrix}$.

Answer.

(a)

$$\left[\begin{array}{cc|c} 3 & 2 & 5 \\ 6 & 3 & 7 \end{array}\right] \xrightarrow[r_1 \to \frac{1}{3} r_1]{} \left[\begin{array}{cc|c} 1 & \frac{2}{3} & \frac{5}{3} \\ 6 & 3 & 7 \end{array}\right] \xrightarrow[r_2 \to r_2 - 6r_1]{} \left[\begin{array}{cc|c} 1 & \frac{2}{3} & \frac{5}{3} \\ 0 & -1 & -3 \end{array}\right]$$

$$\xrightarrow[r_2 \to -r_2]{} \left[\begin{array}{cc|c} 1 & \frac{2}{3} & \frac{5}{3} \\ 0 & 1 & 3 \end{array}\right] \xrightarrow[r_1 \to r_1 - \frac{2}{3} r_2]{} \left[\begin{array}{cc|c} 1 & 0 & \frac{-1}{3} \\ 0 & 1 & 3 \end{array}\right].$$

$$x_1 = \frac{-1}{3}, \qquad x_2 = 3.$$

The system is consistent and has a unique solution $x^* = \begin{pmatrix} \frac{-1}{3} \\ 3 \end{pmatrix}$.

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(b)

$$\left[\begin{array}{ccc|c} 3 & 2 & 4 & 5 \\ 1 & 1 & -3 & 2 \\ 4 & 3 & 1 & 7 \end{array}\right] \xrightarrow[r_1 \leftrightarrow r_2]{} \left[\begin{array}{ccc|c} 1 & 1 & -3 & 2 \\ 3 & 2 & 4 & 5 \\ 4 & 3 & 1 & 7 \end{array}\right] \xrightarrow[r_2 \to r_2 - 3r_1]{r_3 \to r_3 - 4r_1} \left[\begin{array}{ccc|c} 1 & 1 & -3 & 2 \\ 0 & -1 & 13 & -1 \\ 0 & -1 & 13 & -1 \end{array}\right]$$

$$\xrightarrow[r_2 \to -r_2]{} \left[\begin{array}{ccc|c} 1 & 1 & -3 & 2 \\ 0 & 1 & -13 & 1 \\ 0 & -1 & 13 & -1 \end{array}\right] \xrightarrow[r_3 \to r_3 + r_2]{} \left[\begin{array}{ccc|c} 1 & 1 & -3 & 2 \\ 0 & 1 & -13 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow[r_1 \to r_1 - r_2]{} \left[\begin{array}{ccc|c} 1 & 0 & 10 & 1 \\ 0 & 1 & -13 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

$$x_1 + 10x_3 = 1, \qquad x_2 - 13x_3 = 1, \qquad x_3 \text{ free.}$$

$$\boxed{x = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -10 \\ 13 \\ 1 \end{pmatrix}, \qquad t \in \mathbb{R}.}$$

The system is consistent and has infinitely many solutions.

(c)

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & -1 & 2 & 1 \\ 3 & 0 & 5 & 3 \end{array}\right] \xrightarrow[r_2 \to r_2 - r_1]{} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -3 & 1 & 1 \\ 3 & 0 & 5 & 3 \end{array}\right] \xrightarrow[r_3 \to r_3 - 3r_1]{} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -3 & 1 & 1 \\ 0 & -6 & 2 & 3 \end{array}\right] \xrightarrow[r_3 \to r_3 - 2r_2]{} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -3 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

The last row corresponds to the equation $0x_1 + 0x_2 + 0x_3 = 1$, which is impossible. -

Hence the system is inconsistent and has no solution.

(d)

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 1 & 1 & 1 & 3 \\ 3 & 4 & 2 & 4 \end{array}\right] \xrightarrow[r_1 \leftrightarrow r_2]{} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 3 & 1 & 1 \\ 3 & 4 & 2 & 4 \end{array}\right] \xrightarrow[r_2 \to r_2 - 2r_1]{r_3 \to r_3 - 3r_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & -5 \\ 0 & 1 & -1 & -5 \end{array}\right] \xrightarrow[r_3 \to r_3 - r_2]{} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & -5 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow[r_1 \to r_1 - r_2]{} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 8 \\ 0 & 1 & -1 & -5 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$x_1 + 2x_3 = 8, \quad x_2 - x_3 = -5, \quad x_3 \text{ free.}$$

$$\boxed{x = \begin{pmatrix} 8 \\ -5 \\ 0 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.}$$

The system is consistent and has infinitely many solutions (one free parameter t).

(e)

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 11 \\ 2 & 1 & -1 & 1 \\ 1 & -1 & 3 & 10 \end{array}\right] \xrightarrow[r_2 \to r_2 - 2r_1]{r_3 \to r_3 - r_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 11 \\ 0 & -3 & -3 & -21 \\ 0 & -3 & 2 & -1 \end{array}\right] \xrightarrow[r_2 \to -r_2]{} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 11 \\ 0 & 3 & 3 & 21 \\ 0 & -3 & 2 & -1 \end{array}\right]$$

$$\xrightarrow[r_3 \to r_3 + r_2]{} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 11 \\ 0 & 3 & 3 & 21 \\ 0 & 0 & 5 & 20 \end{array}\right] \xrightarrow[r_2 \to r_2 - \frac{3}{5}r_3]{r_1 \to r_1 - \frac{1}{5}r_3} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 7 \\ 0 & 3 & 0 & 8 \\ 0 & 0 & 1 & 4 \end{array}\right] \xrightarrow[r_1 \to r_1 - \frac{2}{3}r_2]{r_2 \to \frac{1}{3}r_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & \frac{8}{3} \\ 0 & 0 & 1 & 4 \end{array}\right]$$

$$x_1 = 1, \quad x_2 = \frac{8}{3}, \quad x_3 = 4$$

The system is consistent and has a unique solution $x^* = \begin{pmatrix} 1 \\ 8/3 \\ 4 \end{pmatrix}$.

(f)

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 8 \\ 4 & 7 & 5 & 20 \\ 0 & -2 & 2 & 0 \end{array}\right] \xrightarrow[r_2 \to r_2 - 2r_1]{} \left[\begin{array}{ccc|c} 2 & 3 & 1 & 8 \\ 0 & 1 & 3 & 4 \\ 0 & -2 & 2 & 0 \end{array}\right] \xrightarrow[r_3 \to r_3 + 2r_2]{} \left[\begin{array}{ccc|c} 2 & 3 & 1 & 8 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 8 & 8 \end{array}\right]$$

$$\xrightarrow[r_1 \to r_1 - 3r_2]{r_3 \to r_3/8} \left[\begin{array}{ccc|c} 2 & 0 & -8 & -4 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & 1 \end{array}\right] \xrightarrow[r_1 \to r_1 + 8r_3]{r_2 \to r_2 - 3r_3} \left[\begin{array}{ccc|c} 2 & 0 & 0 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right] \xrightarrow[r_1 \to r_1/2]{} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$$x_1 = 2, \quad x_2 = 1, \quad x_3 = 1$$

$$\text{The system is consistent and has a unique solution } x^* = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

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3. Suppose the matrix $A \in \mathbb{R}^{4\times 4}$ is defined as the product of three squared matrices, i.e.,

$$A = BCD = \begin{bmatrix} 1 & 3 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 1 & -1 & 2 & 1 \end{bmatrix}.$$

Solve $Ax = \begin{bmatrix} 12 & 6 & 32 & 2 \end{bmatrix}^T$ for $x \in \mathbb{R}^4$.

Answer.

$$Ax = b \quad \Rightarrow \quad BCDx = b \quad \Rightarrow \quad Dx = z, \quad Cz = y, \quad By = b$$

Step 1: Solve $By = b$

$$\begin{bmatrix} 1 & 3 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 12 \\ 6 \\ 32 \\ 2 \end{bmatrix} \quad \Rightarrow \quad y = \begin{bmatrix} 4 \\ 2 \\ 30 \\ 2 \end{bmatrix}$$

Step 2: Solve $Cz = y$

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 30 \\ 2 \end{bmatrix} \quad \Rightarrow \quad z = \begin{bmatrix} -4 \\ 1 \\ -10 \\ -1 \end{bmatrix}$$

Step 3: Solve $Dx = z$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 1 & -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \\ -10 \\ -1 \end{bmatrix} \quad \Rightarrow \quad x = \begin{bmatrix} -4 \\ 9 \\ -23 \\ 58 \end{bmatrix}$$

$$\text{The system is consistent and has a unique solution } x^* = \begin{pmatrix} -4 \\ 9 \\ -23 \\ 58 \end{pmatrix}.$$

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4. Computers are really efficient in doing matrix operations. This is the reason we convert mathematical operations into matrix operations. We are interested to convert operations like *scaling row, sum of two rows and swap two rows* into multiplication of a squared matrix with augmented matrix, i.e. $P * A$.
Considering augmented matrix A,

$$A = \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array}\right].$$

Find the matrix P for the following operations:

(a) $r_1 \leftrightarrow r_2$

(b) $r_3 \leftrightarrow r_2$

(c) $r_1 + r_2 \rightarrow r_1$

(d) $cr_1 \rightarrow r_1$

(e) $ar_2 + br_3 \rightarrow r_2$

Answer.

(a) $r_1 \leftrightarrow r_2$:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) $r_3 \leftrightarrow r_2$:

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

(c) $r_1 + r_2 \to r_1$:

$$P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(d) $cr_1 \to r_1$:

$$P = \begin{bmatrix} c & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(e) $ar_2 + br_3 \to r_2$:

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & a & b \\ 0 & 0 & 1 \end{bmatrix}$$

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5. Write a Python program that:

   (a) asks the user to enter a matrix A (the coefficient matrix) and a vector b (the constant terms on the right-hand side of the equations);

   (b) applies the **Gauss–Jordan elimination algorithm** to solve the linear system $Ax = b$;

   (c) determines and displays whether the system:

   - has a **unique solution** x^*,
   - has **infinitely many solutions**, or
   - has **no solution**.

Answer.

Listing 1: Solving Linear System of Equations via Gauss-Jordan

```python
import numpy as np


# Gauss-Jordan elimination function
def gauss_jordan(A, b):
    A = A.astype(float)
    b = b.astype(float)
    n = len(b)
    aug = np.hstack([A, b.reshape(-1, 1)])
    print("\nInitial augmented matrix:\n", aug, "\n")

    for i in range(n):
        pivot = aug[i, i]
        if np.isclose(pivot, 0):
            for j in range(i+1, n):
                if not np.isclose(aug[j, i], 0):
                    aug[[i, j]] = aug[[j, i]]
                    pivot = aug[i, i]
                    print(f"Swapped row {i+1} with row {j+1}:\n", aug, "\n")
                    break
        if np.isclose(pivot, 0):
            continue
        aug[i] = aug[i] / pivot
        print(f"Scaled row {i+1} to make pivot 1:\n", aug, "\n")
        for j in range(n):
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                if j != i:
                    aug[j] = aug[j] - aug[j, i] * aug[i]
                    print(f"Eliminated element in row {j+1}, column {i+1}:\n", aug, "\n")

    print("Final reduced row-echelon form:\n", aug, "\n")

    for i in range(n):
        if np.all(np.isclose(aug[i, :-1], 0)) and not np.isclose(aug[i, -1], 0):
            print("The system has no solution.")
            return None

    rank = np.sum(~np.all(np.isclose(aug[:, :-1], 0), axis=1))
    if rank < n:
        print("The system has infinitely many solutions.")
    else:
        print("The system has a unique solution:")
        x = aug[:, -1]
        print(x)
m = int(input("Enter the number of rows: "))
n = int(input("Enter the number of columns: "))

print("Enter the matrix A row by row, numbers separated by space:")
A = []
for i in range(m):
    row = list(map(float, input(f"Row {i+1}: ").split()))
    A.append(row)
A = np.array(A)

print("Enter the vector b (right-hand side), numbers separated by space:")
b = np.array(list(map(float, input().split())))

gauss_jordan(A, b)
```